Resource-Constrained Neural Monte Carlo Tree Search for Quantum Boolean Oracle Synthesis

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Abstract

Boolean oracle synthesis embeds classical predicates into quantum algorithms, but direct algebraic-normal-form constructions can be costly in non-Clifford gates, CNOTs, logical depth, and temporary workspace. We formulate bit-flip Boolean oracle synthesis as a resource-constrained search problem over ANF/FPRM term sets and introduce Resource-NMCTS, a logical-layer synthesis framework combining neural action priors, Monte Carlo tree search, Boolean-ring linear-factor screening, depth-frontier policies, Pareto archives, and baseline-preserving guards. The method does not use the SSHR parallelotope candidate space; SSHR is used only as a CNOT-oriented small-function baseline. On 177 traditional functions with $n \leq 6$, Pareto-Resource-NMCTS reduces mean T-count by 72.25% and mean weighted score by 67.80% relative to direct ANF synthesis. Against ESOP beam, weighted ESOP-MILP, and SSHR-H, its score win/loss/tie counts are 174/0/3, 167/3/7, and 173/4/0, respectively. External probes strengthen this comparison. Against a ROS-style LUT proxy over 309 functions, Pareto-Resource-NMCTS obtains 309/0/0 score wins with a mean 84.27% reduction; line-aware and minimum-ancilla reselections of the same verified LUT sweep preserve 309/0/0 score wins with 84.45–85.83% reductions. Against an official-header mockturtle KLUT-to-XAG probe, it obtains 166/11/0 wins on the traditional set and 64/0/0 wins on an $n = 14$ high-dimensional set. A CirKit 3 AIG/multiplicative-complexity probe returns verified rows for 177 traditional and 64 high-dimensional functions; Pareto-Resource-NMCTS wins all score comparisons, with mean reductions of 62.34% and 94.46%, but loses most depth comparisons. A legacy RevKit CLI probe embeds each function as the exact reversible oracle permutation $(x, y) \mapsto (x, y \oplus f(x))$ and runs TBS/DBS/RMS synthesis flows. All 531 direct flow rows return; against the per-function RevKit best-score portfolio, Pareto-Resource-NMCTS has 173/0/4 score wins and a mean score reduction of 67.28%, but uses more peak ancilla on 169/177 functions. In the phase-search branch, a rank-trained diverse policy exact-scores 512 of 8192 affine candidates per held-out $n = 6$ function and matches the wide-search mean within 0.01% under the $T/R_z = 30$ proxy. A sparse depth-2/4 frontier exactly matches the measured depth-2/3/4 frontier on 186 scale and truth-bridge instances while reducing frontier evaluation time by 24.94–28.88%. A learned sparse depth-4 gate preserves this sparse frontier score on a 144-pair multi-seed $n = 24, 28, 32, 40$ audit while reducing its evaluation time by 13.43%. High-dimensional correctness is supported by 5640/5640 item-level symbolic runs, 1512/1512 width-probe symbolic runs, 531/531 exact phase-search rows, 7611/7611 policy-pruning rows, and 400/400 complete truth table-bridge method rows for $n = 21$–25. The contribution is therefore a verified logical-layer search method with strong T-count and weighted-score advantages, not a hardware-mapped or depth-only synthesis claim.

1 Introduction

Given a Boolean function $f : \{0, 1\}^n \to \{0, 1\}$, a bit-flip oracle implements

$$|x\rangle|y\rangle \mapsto |x\rangle|y \oplus f(x)\rangle. \tag{1}$$

Such oracles appear in Grover search, constraint solving, and quantum cryptanalysis. Their cost is not determined only by the number of Boolean operations. In a fault-tolerant or resource-limited setting, the chosen Boolean representation, compute/uncompute order, temporary workspace, and phase realization all affect T-count, CNOT count, logical depth, gate count, and peak ancilla.

Existing synthesis routes provide strong local solutions. ESOP, XAG, LUT and BDD representations reduce different Boolean resources; ROS makes LUT-based oracle synthesis resource-aware; RevKit, mockturtle, CirKit, and ABC provide mature logic and reversible-synthesis toolchains; and SSHR uses parallelotope structures in the Boolean hypercube to target small-scale CNOT-oriented synthesis [6, 5, 4, 15, 1, 13, 12, 17, 11, 10, 9]. These methods motivate the baselines used in this paper, but they also expose a gap: a fixed representation or a single resource objective may miss useful tradeoffs when T-count, CNOTs, depth, gates, and ancilla are optimized together.

We address this gap by treating Boolean oracle synthesis as an explicit search problem. The state is an ANF/FPRM term set; actions are algebraic rewrites that can be expanded back to GF(2) semantics; neural policies and MCTS allocate the search budget; and guard rules ensure that learned candidates do not replace a stronger deterministic baseline. The resulting circuit is checked at the plan level, at the emitted X/CNOT/MCT symbolic level, and, for bridge instances, by full truth table oracle simulation.

The paper makes four contributions. First, it defines Resource-NMCTS as a resource-constrained ANF/FPRM search pipeline rather than as an SSHR extension. Second, it combines Boolean-ring linear-factor screening, neural action priors, MCTS, depth-frontier policies, Pareto archives, and baseline-preserving guards into one logical-layer synthesis workflow. Third, it evaluates the method against ESOP, SSHR, ABC/BDD, ROS-style LUT, mockturtle, CirKit, RevKit CLI, and RevKit phase/Rz baselines on matched functions. Fourth, it separates algorithmic gains from engineering guards through ablations and high-dimensional verification bridges.

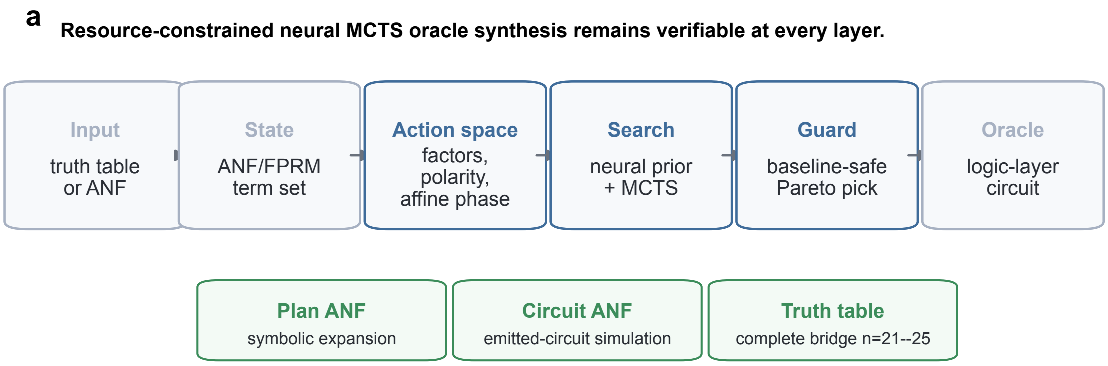

Figure 1: **Verified search pipeline.** Resource-NMCTS operates on complete truth table inputs or ANF term sets, searches algebraic actions under neural and tree-search control, and verifies candidate circuits at plan, emitted symbolic-circuit, and complete-truth table levels.

2 Related Work

2.1 Boolean representations for oracle synthesis

Low-T oracle synthesis is closely tied to the Boolean intermediate representation. BDD-based reversible synthesis uses symbolic decision diagrams to handle large functions [15]. ROS frames oracle synthesis as resource-constrained LUT mapping with downstream reversible implementation costs [6]. XAG-based compilation separates XOR structure from AND nodes, making multiplicative complexity a proxy for non-Clifford cost [5, 4]. Back-end-aware oracle synthesis further connects logic synthesis to downstream resource metrics, but that line includes mapping concerns that are outside the scope of this logical-layer study [16].

2.2 Reversible and logic-synthesis toolchains

ABC supplies mature AIG, XAG/GIA, LUT, and ESOP optimization flows for classical logic [1]. mockturtle and CirKit provide reusable logic-network optimization libraries and shells in the EPFL logic-synthesis ecosystem [13, 10, 9]. RevKit provides reversible-logic and quantum-compilation functionality, including the Python `oracle_synth` API and legacy command-line reversible synthesis flows [12, 11]. In this work these toolchains are used as external baselines or probes. They are not called inside Resource-NMCTS, which keeps the proposed search policy separated from the comparison backends.

2.3 Learning-guided synthesis

Learning-guided synthesis has been explored for both classical and quantum circuits. ShortCircuit uses AlphaZero-style search for classical Boolean circuit design [14]. Gumbel AlphaZero has been used for Clifford+T unitary synthesis [7]; reinforcement learning has been used for non-Clifford-count optimization and ZX-calculus rewriting [2, 8]; MCTS has been used for quantum circuit transformation under connectivity constraints [18]; and MonteQ uses MCTS for Hamiltonian-simulation circuit synthesis [3]. The present work is different in scope: it is a logical-layer Boolean-oracle synthesizer, not a hardware-routing method, not a Hamiltonian-simulation scheduler, not a gate-by-gate unitary generator, and not an SSHR parallelotope search. Its learned component ranks verifiable Boolean-algebraic actions, while ESOP, BDD, XAG/LUT, ROS-style, SSHR, RevKit, mockturtle, CirKit, and ABC flows define the comparison set.

3 Problem and Resource Model

We represent a Boolean function by its square-free ANF

$$f(x) = \bigoplus_{m \in \mathcal{M}} \prod_{i \in m} x_i, \tag{2}$$

where each monomial m is stored as a bit mask. A direct construction applies one controlled action per monomial. The synthesis problem is to replace this direct construction with reusable compute/action/uncompute subplans that implement Eq. (1) while lowering a multi-resource objective.

All resource numbers in this paper are logical-layer estimates over X, CNOT and multi-controlled Toffoli gates. Candidate selection uses

$$\text{score} = T + 0.04\,\text{CNOT} + 0.015\,\text{depth} + 0.01\,\text{gates} + 2\,\text{ancilla}. \tag{3}$$

The score is a ranking objective, not a physical runtime or error model. We therefore report T-count, CNOT count, depth, gate count, and peak ancilla separately whenever a tradeoff is relevant.

4 Method

4.1 Search state and actions

The Resource-NMCTS state is the current set of unsynthesized ANF terms. An action chooses a reusable algebraic structure, synthesizes that structure into a temporary line, uses it to control a quotient subproblem, and then uncomputes it. Candidate actions include common monomial factors, FPRM polarity rewrites, CNOT-computed linear parity factors, Boolean-ring linear factors, and bounded affine changes of variables. Each action is scored by its immediate resource estimate and by the size and structure of the induced subproblems.

4.2 Boolean-ring linear factors

A key extension is allowing the linear factor and the quotient to share variables under Boolean-ring semantics. For example,

$$(x_i \oplus x_j)x_i m = x_i m \oplus x_i x_j m, \tag{4}$$

because $x_i^2 = x_i$ over the Boolean ring. Circuit-wise, the parity $x_i \oplus x_j$ is computed by CNOTs into a temporary line, used as a control for the quotient plan, and uncomputed. This expands the algebraic action space without changing the GF(2) verifier.

4.3 Neural prior, MCTS, frontier policy, and guard

The neural action prior ranks candidate actions using term-count, degree, coverage, variable-support, and immediate-resource features. MCTS then explores combinations of actions with deterministic rollout estimates. A separate frontier policy selects among depth-2, depth-3, and depth-4 Boolean screening frontiers; a cost-aware version trades a small score increase for lower planning time and reduced ancillary lifetime area. We also use two conditional frontier controllers. The staged controller evaluates depth-2 and depth-3 screens first and only runs depth-4 when the measured depth-3 improvement is large enough. The sparse depth-4 gate instead uses the depth-2 state to predict whether the deterministic sparse depth-2/4 frontier needs its depth-4 evaluation; its threshold is selected on validation data to avoid false skips of improving depth-4 cases. Finally, a guard compares learned or expensive candidates to deterministic baselines and returns the best valid candidate under Eq. (3). This makes learning a search-control component, not the source of semantic correctness.

4.4 Pareto archive

The Pareto variant evaluates candidates under several internal objectives, including active-score, T-sparse, CNOT/depth-heavy, gate-sensitive, and ancilla-tight rankings. It removes candidates dominated simultaneously in T-count, CNOT count, depth, gate count, and peak ancilla, then selects from the remaining archive using the active score. The archive is used only after each candidate has passed the same plan and emitted-circuit checks as the base method.

4.5 Phase/Rz branch

RevKit `oracle_synth` returns phase/Rz netlists rather than direct bit-flip Clifford+T circuits. To avoid treating that boundary only as a cost sensitivity, we also implement a phase-parity emitter: each ANF monomial is expanded into merged parity-phase gadgets and verified up to global phase. Fixed-polarity FPRM search and bounded affine preconditioning are then added as phase-search actions. This branch is reported as a logical phase/Rz proxy; it does not output approximate rotation sequences or mapped hardware circuits.

5 Experimental Design

Experiments answer four questions. First, does Resource-NMCTS reduce T-count and weighted score on small functions where full truth table checking is practical? Second, do the conclusions survive comparisons with ESOP, SSHR, ABC/BDD, ROS-style LUT, mockturtle, CirKit, RevKit CLI, and RevKit phase/Rz baselines? Third, which components–affine search, MCTS, neural priors, guards, frontier policies, and Pareto archives–account for the gains? Fourth, how far can correctness evidence be pushed beyond small truth tables?

The traditional slice contains 177 functions with $n \leq 6$. High-dimensional logic-network probes include exported $n = 14$, $n = 15$, $n = 16$, and $n = 18$ random-ANF sets for ABC/BDD, and $n = 14$ sets for mockturtle and CirKit. Scale and bridge experiments cover item-level $n = 20$–40 term sets and complete truth table bridge functions at $n = 21$–25. Paired comparisons include only matching functions or items with successful semantic checks and no timeout. Table 1 summarizes the comparison evidence and the claim boundary for each baseline family.

Table 1: Reviewer-facing evidence matrix. The table consolidates the baseline family, coverage, verification count, headline paired result, and scope boundary used by the manuscript claims.

Evidence block	Scope	Verified rows	Main score result	Boundary
Internal logical baselines	$n = 3$–6	1770/1770	Pareto vs direct ANF 172/1/4, -67.80%; vs ESOP-MILP 167/3/7, -29.84%	Same verifier/cost model; includes direct, AND-direct, ESOP beam/MILP, SSHR-H, MCTS, affine, and Pareto variants.
Exported SSHR/ABC/BDD extension	$n = 3$–6	1416/1416	Resource-NMCTS vs SSHR-I CNOT 172/5/0, -29.48%; vs ABC-XAG 177/0/0, -64.15%	SSHR-I rows use an 8 s Gurobi budget; ABC/BDD rows are logical estimates over exported truth tables.
ROS-style LUT proxy	$n = 3$–6, 14, 15, 16, 18	309/309 best rows; 927/927 K-sweep rows	Pareto vs best-K proxy 309/0/0, -84.27%	Verified LUT proxy only; no official ROS SAT garbage management, reversible emission, or hardware mapping.
mockturtle KLUT-to-XAG probe	$n = 3$–6, 14	241/241	$n \leq 6$ 166/11/0, -31.50%; $n = 14$ 64/0/0, -91.49%	Official-header XAG resynthesis probe; still a logical proxy, not full ROS or reversible garbage management.
CirKit AIG/MC probe	$n = 3$–6, 14	241/241	$n \leq 6$ 177/0/0, -62.34%; $n = 14$ 64/0/0, -94.46%	AIG multiplicative-complexity probe; strongest current depth-oriented external counterpoint.
Legacy RevKit CLI exact oracle	$n = 3$–6	708/708	Pareto vs best-score portfolio 173/0/4, -67.28%	Exact reversible oracle permutation; CNOT/depth are derived from Toffoli-control distributions and ancilla is a visible tradeoff.
RevKit phase/Rz branch	$n = 3$–6	531/531	Affine phase vs RevKit oracle_synth 177/0/0, -65.50%	Logical phase/Rz proxy verified up to global phase; not an approximate-rotation or final Clifford+T sequence.
Learned phase pruning	$n = 3$–6	7611/7611	Held-out n=6 rank-diverse top512 vs wide128 0/7/31, +0.00%	Policy-pruned affine candidate ranking; evidence is held-out exact scoring, not a standalone compiler backend.
High-dimensional frontier search	$n = 20$, 24, 28, 32, 40	1608/1608	Stage-gated vs all-depth scale 0/4/92, +0.04% score; -25.43% time	Term-set symbolic verification with emitted-circuit ANF checks; depth frontier is a planning guard, not a hardware scheduler.
Complete truth-table bridges	$n = 21$–25	400/400	Bridge rows verify plan, emitted symbolic circuit, and complete truth table for $n = 21$–25.	Bridge set is intentionally narrow because complete truth tables grow exponentially.

The experiments were executed as a reproducible artifact package rather than as isolated hand runs. Table 2 records the local compute envelope, manifest-level parallelism, artifact counts, and external toolchain provenance. Runtime numbers should be read within this workstation context; the logical resource counts are the portable experimental outcome.

Table 2: Compute and reproducibility audit. The table documents the workstation, parallel-run manifests, artifact coverage, and external-tool commits used by the current submission package.

Aspect	Evidence	Boundary
Host compute envelope	Apple M4 Pro; 14 physical/14 logical CPU cores; 24.0 GiB RAM; Apple M4 Pro, 20 GPU cores, Metal 4; Python 3.11.15	Workstation context for reproducibility; not a hardware-mapping result.
Learning accelerator	PyTorch 2.12.0; MPS=True; CUDA=False	GPU/MPS is available for neural training; synthesis resources remain logical-layer estimates.
Parallel-run provenance	54 manifests record worker counts; max workers=10; traditional resource 10 workers/1770 rows; RevKit CLI 8 workers/708 rows; ROS-style LUT 8 workers/927 rows; CirKit 8 workers/177 rows; mockturtle 4 workers/177 rows	Wall times are machine-dependent and are not claimed as portable speedups.
Artifact coverage	47 top-level run/train/analyze scripts; 142 raw CSVs; 153 summaries; 65 manifests; 138 paper tables; 5 figure panels (15 PDF/PNG/SVG assets)	Counts describe the current artifact package, not independent datasets.
External-tool provenance	mockturtle 25beb0e; CirKit 4531533; RevKit CLI 104eb35	Toolchain rows are logic-level probes or exact-oracle reversible probes, not full hardware flows.

6 Results

6.1 Traditional functions

Figure 2 and Table 3 summarize the traditional $n \leq 6$ slice. Pareto-Resource-NMCTS has the lowest mean T-count and weighted score among the shown internal baselines, reducing T-count by 72.25% and score by 67.80% relative to direct ANF synthesis. SSHR-H retains the lowest mean CNOT count, so the result is not a CNOT-only dominance claim.

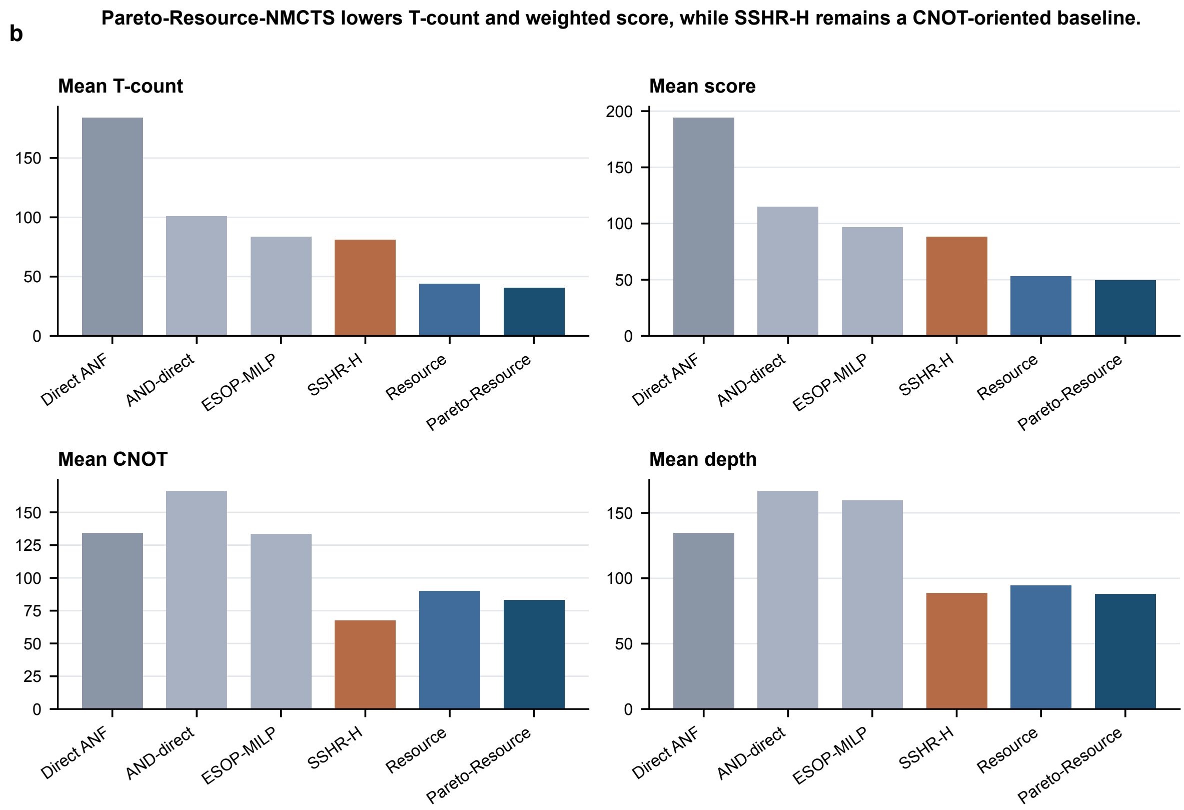

Figure 2: **Mean logical resources on the traditional slice.** Pareto-Resource-NMCTS has the lowest T-count and weighted score; SSHR-H remains a strong CNOT-oriented baseline.

Table 3: Mean logical resources on the $n \leq 6$ traditional slice.

Method	Mean T	Mean CNOT	Mean depth	Mean ancilla	Mean score
Direct ANF	184.1	134.2	134.6	1.33	194.3
AND-direct ANF	101.0	166.5	166.9	2.18	115.2
ESOP-MILP	83.6	133.5	159.6	2.32	96.7
SSHR-H	81.0	**67.6**	88.7	1.36	88.2
Resource-NMCTS	43.9	89.9	94.5	1.92	53.2
Pareto-Resource-NMCTS	**40.4**	83.0	**88.0**	2.03	**49.6**

6.2 External logic-network and reversible-synthesis probes

Figure 3 summarizes paired score changes against several external or traditional baselines. The ROS-style LUT proxy covers 309 functions and 927 K-sweep rows, all truth-table checked; Pareto-Resource-NMCTS wins 309/309 score comparisons against the best-K proxy. The mockturtle probe uses official headers to resynthesize KLUT networks into XAGs; Pareto-Resource-NMCTS has 166/11/0 score wins on the traditional set and 64/0/0 on the $n = 14$ set.

We further reselect the same verified LUT sweep under stricter line pressure. The minimum-ancilla selector reduces the LUT proxy's peak ancilla by 32.47% relative to the best-score proxy, but increases its own weighted score by 40.67%. Pareto-Resource-NMCTS still obtains 309/0/0 score wins against this minimum-ancilla selector and against two line-weighted selectors. This result is a robustness check for LUT-parameter choice, not a claim that the official ROS SAT garbage-management flow has been reproduced.

Table 4: Line-aware reselection of the verified ROS-style LUT proxy sweep.

Comparison	Metric	Items	W/L/T	Mean Δ
and_pareto_resource_nmcts vs external_ros_lut_min_ancilla	score	309	309/0/0	-85.83%
and_pareto_resource_nmcts vs external_ros_lut_min_ancilla	peak_ancilla	309	301/0/8	-68.21%
and_pareto_resource_nmcts vs external_ros_lut_line_w10	score	309	309/0/0	-84.45%
and_pareto_resource_nmcts vs external_ros_lut_k4_line_cap	score	309	309/0/0	-85.02%
external_ros_lut_min_ancilla vs external_ros_lut_proxy	peak_ancilla	309	197/0/112	-32.47%
external_ros_lut_min_ancilla vs external_ros_lut_proxy	score	309	0/197/112	+40.67%

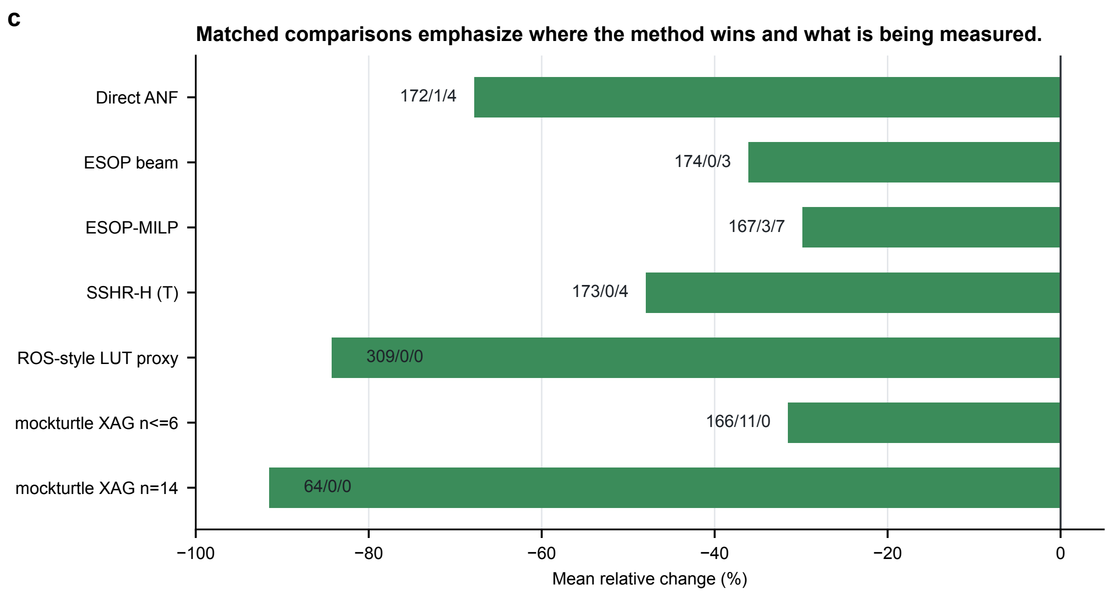

Figure 3: **Paired baseline comparisons.** Negative values favor the target method. Labels show win/loss/tie counts.

The CirKit probe converts the same BLIF benchmarks to AIGER with ABC, applies CirKit AIG rewriting and multiplicative-complexity analysis, writes Verilog, and verifies the Verilog readback through ABC. It produces 177/177 correct traditional rows and 64/64 correct $n = 14$ rows. Pareto-Resource-NMCTS wins all score comparisons but loses depth on most rows, making CirKit the strongest depth-oriented external probe in the current manuscript.

Table 5: CirKit 3 AIG/multiplicative-complexity probe on $n \leq 6$ functions.

Target vs CirKit AIG/MC	Items	W/L/T	Mean Δ
and_resource_nmcts	177	177/0/0	-60.84%
and_pareto_resource_nmcts	177	177/0/0	-62.34%
and_fprm_polarity_archive	177	177/0/0	-58.96%
and_direct_anf	177	124/53/0	-16.74%
direct_anf	177	46/131/0	+35.23%
and_esop_milp	177	150/27/0	-39.41%
sshr_h	177	164/13/0	-32.83%

Table 6: CirKit 3 AIG/multiplicative-complexity probe on the $n = 14$ set.

Target vs CirKit AIG/MC	Items	W/L/T	Mean Δ
and_resource_nmcts	64	64/0/0	-94.42%
and_profile_resource_nmcts	64	64/0/0	-94.42%
and_pareto_resource_nmcts	64	64/0/0	-94.46%
and_fprm_linear_pair	64	64/0/0	-94.27%
and_fprm_root_beam	64	64/0/0	-94.11%
and_direct_anf	64	64/0/0	-91.76%
direct_anf	64	64/0/0	-86.22%
and_fprm_greedy	64	64/0/0	-94.08%
and_affine_greedy	64	64/0/0	-94.08%

The RevKit CLI probe supplies a closer reversible-synthesis comparison. Each function is embedded as the exact oracle permutation $(x, y) \mapsto (x, y \oplus f(x))$, and legacy RevKit reads the SPEC permutation before running TBS, DBS, and RMS synthesis. All 531 direct flow rows return. The per-function best-score RevKit portfolio is strong on line count, but Pareto-Resource-NMCTS retains 173/0/4 score wins and a 67.28% mean score reduction.

Table 7: Legacy RevKit CLI reversible-synthesis portfolio on $n \leq 6$ functions. Each function uses the best-score result among TBS, DBS, and RMS.

Target vs RevKit CLI best-score	Items	W/L/T	Mean Δ
and_resource_nmcts	177	173/0/4	-66.32%
and_pareto_resource_nmcts	177	173/0/4	-67.28%
and_fprm_polarity_archive	177	173/0/4	-64.24%
and_direct_anf	177	167/6/4	-32.79%
direct_anf	177	87/86/4	+5.78%
and_esop_milp	177	172/1/4	-51.50%
sshr_h	177	170/7/0	+83.84%

6.3 Paired statistical evidence

The headline reductions above are paired by function or item, but mean relative improvements alone can hide skewed instances. We therefore recompute the central score comparisons directly from usable raw rows and apply an exact two-sided sign test that ignores ties. Table 8 shows that the main traditional, external-toolchain, and high-dimensional score claims have negative median relative changes as well as strongly one-sided win counts. The high-dimensional root-beam and fast-pair margins are smaller in magnitude than the external-probe margins, but they remain consistent across paired rows.

Table 8: Paired statistical evidence for score comparisons. Rows are recomputed from correct, non-skipped raw CSV entries matched by item name. Negative mean and median changes favor the target method; $\log_{10} p$ is from a two-sided exact sign test over wins and losses, ignoring ties.

Comparison	Pairs	W/L/T	Mean Δ	Median Δ	$\log_{10} p$
$n \leq 6$ Pareto vs direct ANF	177	172/1/4	−67.80%	−71.37%	-49.54
$n \leq 6$ Pareto vs ESOP beam	177	174/0/3	−36.09%	−32.56%	-52.08
$n \leq 6$ Pareto vs ESOP-MILP	177	167/3/7	−29.84%	−23.69%	-44.96
$n \leq 6$ Pareto vs SSHR-H	177	173/4/0	−41.06%	−43.05%	-45.37
$n \leq 6$ Pareto vs SSHR-I CNOT	177	173/4/0	−31.89%	−33.91%	-45.37
$n \leq 6$ Pareto vs ABC-XAG	177	177/0/0	−65.58%	−65.25%	-52.98
ROS-style LUT best-K	309	309/0/0	−84.27%	−81.38%	-92.72
mockturtle XAG $n \leq 6$	177	166/11/0	−31.50%	−32.43%	-35.96
mockturtle XAG $n = 14$	64	64/0/0	−91.49%	−95.50%	-18.96
CirKit AIG/MC $n \leq 6$	177	177/0/0	−62.34%	−62.35%	-52.98
CirKit AIG/MC $n = 14$	64	64/0/0	−94.46%	−96.10%	-18.96
RevKit CLI exact oracle	177	173/0/4	−67.28%	−70.18%	-51.78
$n = 14$ Pareto vs root beam	64	60/0/4	−6.31%	−4.68%	-17.76
$n = 16$ Resource vs root beam	24	23/0/1	−4.36%	−3.36%	-6.62
$n = 18$ Resource vs fast pair	12	12/0/0	−3.55%	−1.79%	-3.31

6.4 Raw multi-resource dominance

Because Eq. (3) is only a ranking objective, we also test dominance without using the weighted score. A method is said to dominate another row only if it is no worse in T-count, CNOT count, logical depth, and peak ancilla, and is strictly better in at least one of them. Table 9 shows the resulting four-resource dominance counts for Pareto-Resource-NMCTS on the traditional slice. The method strongly dominates ESOP beam and many ESOP-MILP rows, but most comparisons against SSHR, CirKit, mockturtle, and RevKit are incomparable. This confirms that those baselines occupy real CNOT, depth, or line-count tradeoff points even when their weighted scores are worse.

Table 9: Raw four-resource dominance of Pareto-Resource-NMCTS on $n \leq 6$ functions. Dominance uses only T-count, CNOT count, logical depth, and peak ancilla; weighted score is not part of the predicate.

Baseline	Pairs	Pareto dom.	Base dom.	Incomp.	Equal
Direct ANF	177	69	1	103	4
ESOP beam	177	165	0	9	3
ESOP-MILP	177	123	2	45	7
SSHR-H	177	33	4	140	0
SSHR-I CNOT	177	9	3	165	0
ABC-XAG	177	33	0	144	0
mockturtle XAG	177	11	0	166	0
CirKit AIG/MC	177	21	0	156	0
RevKit CLI exact oracle	177	3	0	170	4

The same predicate can be applied to the whole method pool. Table 10 counts how often each method remains non-dominated by any other listed method on the same function. Pareto-Resource-NMCTS is non-dominated on 170/177 functions, while SSHR-I, mockturtle, and RevKit remain frequently non-dominated because they keep distinct resource tradeoffs.

Table 10: Non-dominated counts in the 12-method $n \leq 6$ pool.

Method	Rows	Non-dominated	Dominated
Pareto-Resource-NMCTS	177	170	7
SSHR-I CNOT	177	162	15
mockturtle XAG	177	159	18
RevKit CLI exact oracle	177	141	36
Resource-NMCTS	177	124	53
CirKit AIG/MC	177	56	121
SSHR-H	177	46	131
ESOP-MILP	177	44	133
ABC-XAG	177	15	162
Direct ANF	177	7	170
ESOP beam	177	6	171
AND-direct ANF	177	4	173

6.5 Search contribution and ablation

Table 11 consolidates the contribution analysis. The largest pre-portfolio gain comes from affine and Boolean-ring structural search. The learned prior gives smaller but non-degrading quality gains on the traditional slice, while the frontier policies mainly control the quality/time frontier in high-dimensional term-set experiments. Table 12 isolates the high-dimensional Boolean-ring and Boolean-screen branch. This table is deliberately mixed: it shows quality-seeking structural guards, a screen gate that preserves the selected resource vector while saving time, and a negative control where a fast screen is not resource-competitive. This prevents the structural-search claim from being reduced to a neural-prior claim or overstated as a universal screen improvement. Table 13 then compresses the expensive depth-frontier itself. On the listed scale and truth-bridge slices, evaluating only depth 2 and depth 4 exactly matches the full depth-2/3/4 frontier while removing 24.94–28.88% of frontier evaluation time. Table 14 adds a learned gate on top of this sparse frontier. Trained on generated $n = 16, 20, 24$ term sets and applied unchanged to held-out $n = 28, 40$ term sets, it runs depth 4 on 32/48 test pairs, records zero false skips, exactly matches the sparse-frontier score, and reduces sparse-frontier evaluation time by 17.39%. The deterministic sparse frontier remains the quality reference; the learned component controls search budget after the depth-2 screen. Table 15 then audits the same trained gate without retraining on three independent seeds over $n = 24, 28, 32, 40$. Across 144 additional pairs it records zero false skips, matches the deterministic sparse frontier on every row, and saves 13.43% of sparse-frontier evaluation time. Table 16 adds the earlier staged teacher/guard. Its depth-4 trigger is selected only on the large-policy validation split; on the independent $n = 24, 28, 32, 40$ scale rows it remains within 0.04% of all-depth score while reducing staged planning time by 25.43%.

Table 11: Matched contribution decomposition. Negative mean score changes favor the first method in each comparison.

Component	Dataset	Pairs	Score W/L/T	Mean Δ score
Affine greedy vs fixed-coordinate MCTS	ablation_affine	321	165/88/68	−12.12%
Neural refine over affine greedy	ablation_affine	322	65/0/257	−1.08%
Guarded Affine-NMCTS over no-guard	ablation_affine	322	88/0/234	−1.74%
Learned prior for Resource-NMCTS	traditional_resource	177	39/0/138	−1.10%
Pareto archive over Resource-NMCTS	traditional_resource	177	68/0/109	−3.26%
No-MCTS portfolio over heuristic-only	trad. ablation	177	54/0/123	−2.51%
Resource-NMCTS over no-MCTS portfolio	trad. ablation	177	54/0/123	−1.44%
Pareto Resource-NMCTS over no-MCTS portfolio	trad. ablation	177	106/0/71	−4.69%
Highdim no-MCTS portfolio over root beam	n14 ablation	16	14/0/2	−6.25%
Highdim no-MCTS portfolio over linear-pair	n14 ablation	16	14/0/2	−3.08%
Linear-pair guard vs root beam at n=14	highdim_resource	64	60/0/4	−3.00%
Recursive linear-pair guard vs root beam at n=15	n15 scale	32	30/0/2	−5.28%
Recursive linear-pair guard vs shallow linear-pair at n=16	n16 scale	24	23/0/1	−2.54%
Recursive linear-pair guard vs root beam at n=16	n16 scale	24	23/0/1	−4.31%
Baseline-preserving AI guard vs deterministic recursive guard at n=16	n16 scale	24	6/0/18	−0.05%
Fast linear-pair guard vs root beam at n=18	n18 stress	12	6/0/6	−1.91%
Resource-NMCTS vs fast linear-pair at n=18	n18 stress	12	12/0/0	−3.55%

Table 12: Boolean-ring structural evidence in high-dimensional slices. Negative mean changes favor the first method in each comparison. The $n = 18$ row is kept as a speed-only negative control rather than promoted as a quality result.

Slice	Comparison	Pairs	Score W/L/T	Mean Δ score	Mean Δ time	Interpretation
$n = 14$	Boolean guard vs pairwise-wide	12	9/0/3	-0.82%	+32.80%	quality gain; slower exhaustive guard
$n = 16$	Boolean guard vs pairwise-wide	24	14/0/10	-0.34%	+22.80%	quality gain; slower guard
$n = 16$	Boolean guard vs deterministic deep	24	18/0/6	-0.52%	+356.64%	stronger quality branch; expensive
$n = 16$	Boolean-linear deep vs pairwise-wide	24	14/6/4	-0.22%	-72.31%	fast structural variant with small quality gain
$n = 20$	Resource update vs Boolean screen	6	5/0/1	-4.52%	+453.05%	quality frontier; much slower
$n = 20$	Screen gate vs full Resource	6	0/0/6	+0.00%	-75.58%	safe skip: same resources, less planning time
$n = 20$	Recursive Boolean screen vs old Resource	6	5/0/1	-7.47%	-7.32%	large-n positive screen; ancilla tradeoff
$n = 18$	Recursive Boolean screen vs old Resource	12	0/11/1	+36.94%	-97.90%	speed-only negative control; not promoted

Table 13: Sparse depth-frontier audit. The sparse controller evaluates depth-2 and depth-4 Boolean-ring screens, skips depth 3, and is compared with the full measured depth-2/3/4 frontier in the same raw files.

Source	n scope	Pairs	Score W/L/T	Mean Δ score	Mean Δ time	Depth-4 vs depth-3
scale-frontier	20,28,40	72	0/0/72	+0.00%	-27.57%	44/0/28
scale-generalization	24,28,32,40	96	0/0/96	+0.00%	-28.17%	52/0/44
truth-bridge-n23	23	6	0/0/6	+0.00%	-24.94%	5/0/1
truth-bridge-n24	24	6	0/0/6	+0.00%	-28.88%	3/0/3
truth-bridge-n25	25	6	0/0/6	+0.00%	-27.06%	4/0/2

Table 14: Learned sparse depth-4 gate. The gate predicts whether the depth-4 Boolean-ring screen should run after the depth-2 sparse-frontier state has been evaluated. The threshold is selected on validation data with a conservative false-skip guard and is applied unchanged to held-out $n = 28, 40$ test rows.

Split	Pairs	Run d4	False skips	Score W/L/T vs sparse	Mean Δ score	Mean Δ time	Score W/L/T vs depth-2
train	96	78	0	0/0/96	+0.00%	-10.17%	75/0/21
valid	48	38	0	0/0/48	+0.00%	-12.01%	34/0/14
test	48	32	0	0/0/48	+0.00%	-17.39%	23/0/25

Table 15: Independent-seed sparse depth-4 gate audit. The trained gate and its validation-selected threshold are reused without retraining on three new seeds. The table reports aggregate and per-n comparisons against the deterministic sparse depth-2/4 frontier.

Scope	Pairs	Run d4	False skips	Score W/L/T vs sparse	Mean Δ score	Mean Δ time	Score W/L/T vs depth-2
all	144	106	0	0/0/144	+0.00%	-13.43%	94/0/50
n=24	36	29	0	0/0/36	+0.00%	-9.88%	27/0/9
n=28	36	27	0	0/0/36	+0.00%	-12.98%	24/0/12
n=32	36	23	0	0/0/36	+0.00%	-18.19%	21/0/15
n=40	36	27	0	0/0/36	+0.00%	-12.66%	22/0/14

Table 16: Stage-gated depth-frontier controller. The depth-4 trigger is chosen on validation data and applied unchanged to scale and truth-bridge rows.

Source	Baseline	Pairs	Score W/L/T	Mean Δ score	Mean Δ time	Run depth-4
scale_generalization	adaptive_all_depth	96	0/4/92	+0.04%	-25.43%	52
scale_generalization	screen_depth2	96	58/0/38	-2.40%	+893.25%	52
scale_generalization	depth_frontier_policy	96	5/3/88	-0.06%	+101.10%	52
truth_bridge_n23	adaptive_all_depth	6	0/0/6	+0.00%	-11.51%	5
truth_bridge_n23	screen_depth2	6	5/0/1	-2.47%	+1322.09%	5
truth_bridge_n23	depth_frontier_policy	6	1/0/5	-0.12%	+94.20%	5

Table 17 separates promoted learned-control components from limited diagnostics. The frontier policies, sparse depth-4 gate, and phase shortlist provide the paper-facing learned-control evidence. In contrast, the current Boolean-neural guard and root-action neural ranker are kept as bounded diagnostics: they show small quality or evaluation-cost signals, but not enough quality/runtime advantage to carry the main claim.

Table 17: Learned-control evidence audit. Rows marked as limited are reported to bound the role of AI rather than promoted as headline improvements.

Component	Scope	Quality evidence	Cost evidence	Paper role
Depth-frontier policy	held-out $n = 28, 40$; 48 rows	vs oracle frontier 0/3/45, +0.04%	-51.30% time vs all-depth frontier	promoted quality/time selector
Stage-gated frontier	independent $n = 24, 28, 32, 40$; 96 rows	vs all-depth 0/4/92, +0.04%	-25.43% staged planning time	promoted validation-calibrated guard
Sparse depth-4 gate	multi-seed $n = 24, 28, 32, 40$; 144 pairs	vs sparse frontier 0/0/144, +0.00%; false skips 0	-13.43% time vs sparse frontier	promoted budget gate after depth-2
Rank-diverse phase shortlist	held-out $n = 6$ phase search; 38 rows	vs budget-32 17/0/21, -2.48%; vs wide-128 0/7/31, +0.00%	512/8192 exact forms per function	promoted phase-search pruning
Boolean neural guard	$n = 16$ high-dimensional guard; 24 rows	vs deterministic 4/0/20, -0.12%	+94.49% runtime	limited quality guard, not runtime claim
Root-action neural ranker	$n = 14$ root-action diagnostic; 10 rows	vs beam4 +0.03%; oracle24 headroom -0.12%	-98.06% ranking time vs beam4 eval	not promoted; future root-ranker target

6.6 Score-weight robustness

The active score in Eq. (3) is intentionally a logical-resource ranking objective rather than a physical cost model. To check that the main comparisons are not artifacts of one coefficient choice, we recompute scores post hoc on verified rows under T-only, CNOT/depth-heavy, and ancilla-tight profiles. Table 18 shows that the small-function advantages over ESOP and SSHR-H and the high-dimensional advantages over root-beam or fast-pair guards survive these alternative profiles, while the CNOT/depth and ancilla-tight profiles expose the expected narrower margins.

Table 18: Post-hoc robustness to alternative logical-resource weights. Each cell reports score win/loss/tie counts and mean relative score change; negative values favor Resource-NMCTS or Pareto-Resource-NMCTS.

Comparison	Paper score	T-only	CNOT-depth	Ancilla-tight
$n \leq 6$ traditional: ESOP cube beam	174/0/3, -36.09%	174/0/3, -38.95%	174/0/3, -32.08%	174/0/3, -32.75%
$n \leq 6$ traditional: ESOP-MILP	167/3/7, -29.84%	166/0/11, -32.77%	165/4/8, -25.44%	167/3/7, -26.68%
$n \leq 6$ traditional: SSHR-H	173/4/0, -41.06%	173/0/4, -47.93%	168/9/0, -27.87%	172/5/0, -31.95%
$n = 14$ random ANF: FPRM root beam	60/0/4, -6.31%	60/0/4, -7.65%	60/0/4, -5.35%	56/4/4, -3.28%
$n = 16$ random ANF: FPRM root beam	23/0/1, -4.36%	23/0/1, -4.70%	23/0/1, -4.28%	23/0/1, -3.43%
$n = 18$ random ANF: FPRM root beam	12/0/0, -5.29%	12/0/0, -6.19%	12/0/0, -4.54%	11/1/0, -3.33%
$n = 18$ random ANF: fast pair guard	12/0/0, -3.55%	12/0/0, -3.75%	12/0/0, -3.23%	12/0/0, -3.33%

6.7 Phase/Rz search

RevKit Python `oracle_synth` returns on all 177 traditional functions, but 171 rows contain non-Clifford R_z rotations, with 9242 such rotations in total. We therefore treat it as a phase/Rz lower-bound boundary rather than as an exact Clifford+T T-count comparison. The internal phase-parity emitter, fixed-polarity FPRM search, and affine FPRM phase search turn this boundary into a verifiable search branch.

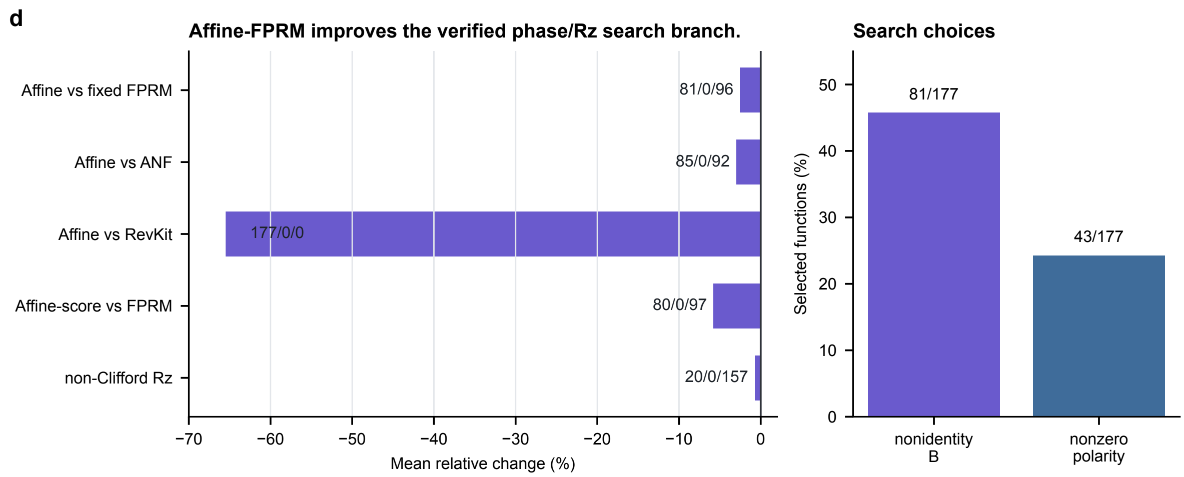

Figure 4: **Affine-FPRM phase search.** Affine preconditioning improves the phase-parity proxy over fixed-polarity FPRM and ANF baselines, while remaining a logical phase/Rz proxy rather than an approximate-rotation sequence.

Table 19: Affine-FPRM phase-parity search comparisons.

Comparison	Items	W/L/T	Mean Δ
Affine-FPRM-$T/R_z = 30$ vs FPRM	177	81/0/96	-2.51%
Affine-FPRM-$T/R_z = 30$ vs ANF	177	85/0/92	-2.98%
Affine-FPRM-$T/R_z = 30$ vs RevKit	177	177/0/0	-65.50%
Affine-FPRM non-Clifford Rz vs FPRM	177	20/0/157	-0.73%
Affine-FPRM-score vs FPRM	177	80/0/97	-5.77%

Increasing the affine transform budget from 32 to 128 yields additional but bounded gains: under the $T/R_z = 30$ proxy, the wide search gives 43/0/134 wins/losses/ties over budget 32 and reduces mean synthesis score by 0.60%. A rank-trained policy and a deterministic diversity reranker then compress this search. On the held-out $n = 6$ split, the diverse top-512 policy exact-scores only 512 of 8192 affine forms per function, keeps the budget-32 comparison at 17/0/21 with a 2.48% mean reduction, and matches the wide-128 mean within 0.01%. Across eight same-budget random repeats, each policy top-k mean is lower than all random-repeat means, although the absolute margins are small. Thus the result supports learned pruned phase-search feasibility, not a claim that the current scorer solves phase/Rz synthesis globally.

Table 20: Affine-FPRM phase-search budget expansion.

Metric	Pairs	W/L/T	Mean Δ
score target	177	38/0/139	−1.59%
score+1/Rz target	177	40/0/137	−0.93%
$T/R_z = 30$ target	177	43/0/134	−0.60%
non-Clifford Rz	177	3/0/174	−0.29%
total Rz	177	35/2/140	−2.39%
CNOT	177	37/2/138	−1.74%
depth	177	41/2/134	−1.93%

Table 21: Rank-trained and diversity-reranked phase candidate pruning on held-out $n = 6$ functions.

Comparison	Items	W/L/T	Mean Δ
Policy top-512 vs budget-32	38	17/0/21	-2.40%
Policy top-512 vs wide-128	38	0/9/29	+0.13%
Diverse policy top-512 vs budget-32	38	17/0/21	-2.48%
Diverse policy top-512 vs wide-128	38	0/7/31	+0.00%
Policy top-64 vs random top-64	38	10/7/21	+1.37%
Diverse policy top-64 vs random top-64	38	10/7/21	+0.56%
Policy top-128 vs random top-128	38	7/6/25	+0.81%
Diverse policy top-128 vs random top-128	38	8/4/26	+0.03%
Policy top-512 total Rz vs budget-32	38	15/0/23	-4.57%
Policy top-512 CNOT vs budget-32	38	16/1/21	-2.80%
Policy top-512 depth vs budget-32	38	16/1/21	-3.29%

6.8 High-dimensional verification

Figure 5 summarizes the current verification envelope. The manuscript-level evidence includes 5640/5640 item-level symbolic runs for $n = 20$–40, 1512/1512 width-probe symbolic runs, 531/531 exact phase-search rows, 7611/7611 phase-policy pruning rows, and 400/400 complete truth table bridge method rows. The complete bridge spans $n = 21$–25 and checks the oracle relation, the expanded ANF plan, and the emitted-circuit symbolic semantics. Table 22 makes the large-scale evidence explicit by separating generated functions or settings from method rows and by reporting representative resource means only for the paper-facing method in each slice.

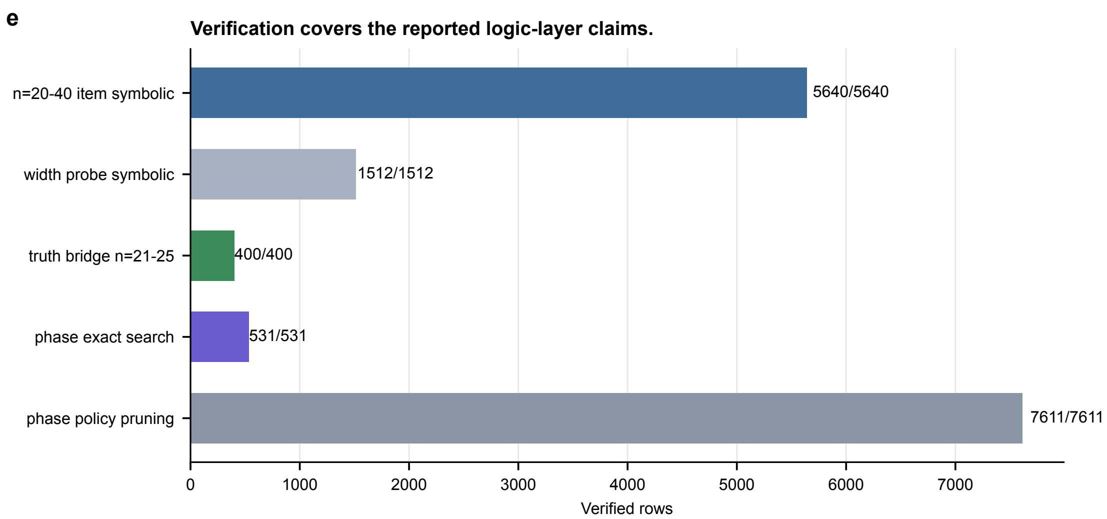

Figure 5: **Verification coverage.** Large term-set experiments are validated symbolically; phase-search and phase-policy rows are checked up to global phase; $n = 21$–25 bridge slices add complete truth table checking on selected generated functions.

Table 22: Scaling and resource audit for high-dimensional experiments. The representative resource mean is method-specific; it is not an average over all baselines in the slice.

Slice	n	Fn./settings	Rows verified	Representative resource mean	Boundary
Large term-set frontier	$24, 28, 32, 40$	96	960/960	score 1090.0; T/CNOT/depth 986/1593/1593; anc. 5.3; time 7.40s	Term-set validation; no full truth table beyond bridge slices.
Stage-gated frontier	$24, 28, 32, 40$	96	96/96	score 1089.7; T/CNOT/depth 985/1593/1593; anc. 5.3; time 10.57s	Controller audit; still term-set symbolic verification.
Action-width probe	$20, 28, 40$	216	1512/1512	score 1130.4; T/CNOT/depth 1023/1648/1648; anc. 5.2; time 1.27s	Sensitivity check; wider root screens are not claimed as better.
Complete truth-table bridge	21–25	40	400/400	score 1126.2; T/CNOT/depth 1019/1647/1647; anc. 5.3; time 4.16s; truth check 0.08s	Complete checking on generated bridge slices only.

Table 23: Verification scope and boundary.

Check	Current coverage	Boundary
$n = 20$–40 item-level symbolic runs	5640/5640	Not full truth-table enumeration.
Width-probe symbolic runs	1512/1512	Supports the default action width, but remains term-set validation.
$n = 21$–25 complete bridge	400/400	Small generated slices; not exhaustive coverage of all high-dimensional functions.
Exact phase-search rows	531/531	Verified up to global phase; no rotation-sequence output.
Phase-policy pruning rows	7611/7611	Learned shortlist rows verified up to global phase; still no rotation-sequence output.

7 Discussion

The evidence supports a specific, bounded claim: Resource-NMCTS is a logical-layer resource-constrained Boolean oracle synthesis method that improves T-count and weighted score over several fixed-representation and external-toolchain baselines. The claim is not that Resource-NMCTS dominates every metric. SSHR remains a strong CNOT-oriented small-function baseline; CirKit AIG/MC remains a strong depth-oriented external probe; and RevKit CLI uses fewer auxiliary lines on most traditional functions.

The ablations also calibrate the role of AI. The neural prior is useful, but the largest gain comes from the searchable algebraic action space and from guarded portfolio selection. This is a better fit for the evidence than claiming that deep reinforcement learning alone drives the result. The Boolean-ring structural audit further separates resource quality from planning cost: at $n = 14$ and $n = 16$ the guard improves score by 0.34–0.82% over the pairwise-wide baseline; at $n = 20$ a deeper screen gives a 7.47% score reduction over the old Resource baseline with an ancilla tradeoff; and the $n = 18$ screen-only row is much faster but worse in score. The sparse depth-frontier audit suggests a simpler planning-cost reduction inside the current frontier: depth 3 was never needed once depth 4 was measured in the audited slices, so a depth-2/4 frontier exactly matched all-depth resources while removing roughly one quarter of frontier evaluation time. This is an empirical controller for the present generated slices, not a proof that depth 3 is universally dominated. The learned sparse depth-4 gate adds a narrower AI-control result: after the sparse frontier has removed depth 3, the gate skips some depth-4 evaluations without changing score in either the held-out split or the independent multi-seed audit under its conservative threshold. The large, cost-aware, sparse-gated, and staged frontier controllers should be read as operating points on the same search frontier: quality-oriented single-shot selection, faster cost-aware selection, learned sparse-budget control, and a validation-calibrated staged teacher. In the phase/Rz branch, affine and polarity search give verified improvements, and the rank-trained diverse shortlist now tracks the wide-search frontier with 6.25% of the exact evaluations. However, random-repeat margins remain small, so this branch should be read as learned pruning evidence rather than as a solved rotation-synthesis policy. The next target is rotation-spectrum-aware optimization and sequence-level audit.

Several limitations are deliberate. All results are logical-layer estimates; the paper does not model routing, hardware connectivity, native gate sets, noise, or magic-state factories. The ROS-style LUT result is a proxy rather than a full ROS reproduction with SAT garbage management; the line-aware LUT reselection only tests parameter sensitivity within the same proxy. The RevKit CLI baseline is a real exact-oracle reversible-synthesis probe, but CNOT/depth are derived from RevKit's Toffoli-control distribution and the experiment is still not a hardware mapping. High-dimensional symbolic checks validate emitted circuit semantics but do not replace full truth-table enumeration beyond the bridge slices.

8 Conclusion

We presented Resource-NMCTS, a neural Monte Carlo tree search framework for resource-constrained quantum Boolean oracle synthesis over verifiable ANF/FPRM actions. On matched small-function benchmarks, the method substantially lowers T-count and weighted score relative to direct ANF, ESOP, SSHR-H, and ESOP-MILP baselines. On external probes based on ROS-style LUT mapping, mockturtle, CirKit, and legacy RevKit CLI reversible synthesis, the method retains a strong weighted-score advantage while exposing clear depth and ancilla tradeoffs. Large symbolic experiments and $n = 21$–25 complete truth-table bridges support the semantic correctness of the logical-layer circuits. The sparse depth-frontier analysis further reduces high-dimensional planning cost by skipping depth 3 in the audited frontier, and the learned sparse depth-4 gate preserves sparse-frontier score across both held-out and independent-seed audits while removing additional depth-4 evaluations. The staged frontier analysis remains a validation-calibrated controller when depth-4 evaluation itself should be conditional. The next step is not to claim hardware-level optimality, but to strengthen the phase/Rz policy and complete fuller toolchain reproductions while keeping the claim at the logical synthesis layer.

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